

Alex Morgan

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Education

University of California, Berkeley

Berkeley, CA

Bachelor of Science in Computer Science & Engineering; GPA: 3.85/4.0

Aug 2020 – May 2024

- **Relevant Coursework:** Distributed Systems, Machine Learning, Operating Systems, Algorithms, Computer Vision.

Technical Skills

Languages: Python, C/C++, Rust, TypeScript, Go, SQL, LaTeX, Bash

Frameworks & Libraries: PyTorch, ROS 2, OpenCV, FastAPI, React, Node.js, NumPy, Docker

Developer Tools: Git, GitHub Actions CI/CD, Linux (Ubuntu), WSL2, Kubernetes, AWS, TensorRT

Specializations: Autonomous Systems, High-Performance Computing, State Estimation, Deep Learning

Experience

Software Engineer Intern

Jun 2023 – Aug 2023

Autonomous Robotics Corp

San Francisco, CA

- Engineered real-time state estimation pipeline in C++ and ROS 2, reducing trajectory drift by **34%** over a 500-meter test circuit.
- Developed automated CI/CD simulation tests with GitHub Actions and Docker, reducing PR regression triage time from 45 min to **8 min**.
- Optimized point cloud registration kernel using CUDA, achieving **65 FPS** on NVIDIA Jetson Orin edge platform.

Undergraduate Research Assistant

Jan 2022 – May 2023

Berkeley AI Research (BAIR) Lab

Berkeley, CA

- Co-authored research on vision-language models for robotic manipulation; evaluated on 10,000+ benchmark trials.
- Implemented distributed training pipelines in PyTorch across 16x A100 GPUs, achieving **2.8x speedup**.

Projects

Latex AI Workflow | Python, LaTeX, TeX Live, GitHub Actions, Docker

Jan 2024

- Architected a local Overleaf alternative with automated visual page inspection and sub-second live compilation.
- Published open-source CLI engine with 100% offline build capability and multi-target GitHub Actions CI.

Distributed Vector Search Engine | Rust, gRPC, SIMD, RocksDB

Sep 2023 – Dec 2023

- Built high-throughput HNSW index vector database serving sub-5ms k-NN queries over 10M embeddings.
- Benchmarked against FAISS; achieved **99.2% recall** with 40% lower memory footprint via scalar quantization.

Publications & Honors

Conference Paper: "Deep Spatial-Temporal Alignment for Robotics Inspection Platforms", IEEE ICRA 2024 (Under Review).

Honors: Dean's Honors List (All Semesters), 1st Place Autonomous Robotics Hackathon 2023.